

SATYAM MISHRA

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Portfolio

PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in developing scalable backend and full-stack applications using **Java, Spring Boot, React.js, PostgreSQL, and REST APIs**. Experienced in building AI-powered applications with **LLM integration, Generative AI, and prompt engineering**, including NL2SQL systems and intelligent automation workflows. Skilled in **secure coding, JWT authentication, API development, SDLC, Agile/Scrum, and cloud-based application development**. Strong foundation in **Data Structures, Algorithms, OOP, and system design fundamentals** with 600+ coding problems solved. Passionate about building scalable, high-performance software systems and backend engineering solutions.

SKILLS

Programming Languages	Java (Core + Advanced), Python, C/C++, JavaScript, Dart, SQL
Data Structures & Algorithms	Arrays, Linked Lists, Trees, Graphs, Dynamic Programming, Recursion, Sorting, Sliding Window, Greedy Algorithms; 600+ DSA problems solved
Backend Engineering	Java, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, RESTful APIs, Microservices Basics, JWT Authentication, API Development, Secure Coding, Object-Oriented Design (OOD), SDLC, Backend System Design
Frontend Development	React.js, HTML5, CSS3, JavaScript, Flutter (Cross-Platform Mobile Development), API Integration
Cloud & Databases	AWS (EC2, S3, IAM basics), Google Cloud (Firebase, Firestore, Cloud Functions), MySQL, PostgreSQL, MongoDB, Firebase Realtime Database
AI / Generative AI	Generative AI, LLM Integration, Prompt Engineering, NLP (NLTK), Pandas, NumPy, Scikit-learn, TensorFlow (basic), AI-powered Automation, NL2SQL Systems
Software Engineering Practices	Agile/Scrum, Git Workflow, Code Reviews, Debugging, Unit Testing (JUnit), API Testing, Performance Optimization, Scalable Application Development, Problem Solving
DevOps & Tools	Git, GitHub, Docker (basic), Postman, Swagger/OpenAPI, pgAdmin, Firebase Console, CI/CD Basics
Automation & IoT	UiPath, Power Automate Desktop, OpenRPA, ESP32, Arduino, Sensors, Actuators, IoT-Cloud Integration

EXPERIENCE

Java Full Stack Developer

September 2025 - Present

HCL Technologies

Pune, Maharashtra

- Developed and maintained scalable backend services and RESTful APIs using **Java, Spring Boot, MySQL, and PostgreSQL**, improving API response performance by **15%** through query optimization, indexing, and efficient connection pooling.
- Owned development and delivery of full-stack web application modules using **Spring Boot**, JavaScript, and React-based frontend integration, contributing 5+ production-ready user stories per Agile/Scrum sprint.
- Implemented secure authentication and authorization workflows using **JWT-based security**, CRUD operations, input validation, and parameterized SQL queries following secure coding and SDLC best practices.
- Collaborated within a 6-member cross-functional Agile team on code reviews, debugging, API integration, sprint planning, and Git/GitHub-based version control, reducing defect leakage by **20%**.
- Designed and documented 10+ REST API endpoints using **Swagger/OpenAPI**, improving developer onboarding, API usability, and integration efficiency across internal modules.
- Contributed to unit testing and application reliability using **JUnit**, maintaining 98%+ test coverage for enterprise backend modules.

Key Project: *Excel AI Chatbot*

Java | Spring Boot | PostgreSQL | LLM | Apache POI | React

- Architected and developed the backend for an **AI-powered Excel Chatbot** enabling users to upload multi-sheet Excel datasets and perform natural language-based querying through a React frontend.
- Built an intelligent Excel parsing and schema inference engine using **Apache POI**, dynamically detecting tables across sheets, generating normalized PostgreSQL schemas, and automating structured data ingestion pipelines.
- Designed and integrated a **Generative AI-powered NL2SQL pipeline** by connecting **LLM APIs** with Spring Boot microservice workflows, converting natural language prompts into parameterized SQL queries executed against PostgreSQL databases.
- Implemented dynamic DDL generation, multi-table relationship mapping, JSON serialization, and REST API-based response handling to ensure scalable and reliable end-to-end conversational data retrieval.
- Improved backend modularity and maintainability using object-oriented design principles, layered architecture, reusable services, and API-driven development practices.

Junior RPA Developer

March 2025 - September 2025

IBOTIX Pvt. Ltd.

Noida, Uttar Pradesh

- Developed and deployed enterprise-grade **Robotic Process Automation (RPA)** workflows using **UiPath, Power Automate, and Python**, automating repetitive business operations and reducing manual processing effort by **40%**.
- Designed secure automation pipelines for credential handling, role-based access workflows, and encrypted data processing while following secure SDLC and compliance-oriented automation practices.
- Built reusable bot components, exception-handling mechanisms, logging frameworks, and scheduler-based execution workflows, improving automation reliability and reducing runtime failures by **25%**.
- Collaborated with cross-functional teams to gather process requirements, perform workflow optimization, debug automation failures, and deliver production-ready bots within Agile-based development cycles.
- Integrated APIs, Excel automation, PDF processing, email automation, and database operations into RPA workflows, enabling seamless end-to-end business process orchestration.
- Utilized Git/GitHub for version control, bot deployment management, and collaborative development, ensuring maintainable and scalable automation solutions.

Key Project: *SecureBotix*

Power Automate | Python | OCR | REST APIs

- Architected and developed **SecureBotix**, a secure intelligent automation platform designed to automate sensitive business workflows involving document validation, credential-based access, and multi-step approval processing.
- Built intelligent document processing pipelines using **OCR**, Excel automation, and PDF data extraction techniques to automate structured and semi-structured data handling with high accuracy.
- Integrated **REST APIs**, email automation, and database-driven workflows into RPA bots, enabling secure real-time data synchronization and automated task execution across enterprise systems.
- Implemented robust exception handling, audit logging, retry mechanisms, and role-based access validations to ensure reliability, traceability, and secure automation execution.
- Improved operational efficiency by automating repetitive manual workflows, reducing turnaround time, minimizing human intervention, and enhancing process scalability for enterprise use cases.

PROJECTS

EcoSentinel: AI-Powered Greenhouse Monitoring SystemIoT | Python | Firebase | Machine Learning

Designed and developed an end-to-end IoT-based smart greenhouse monitoring system using **ESP32 microcontrollers, LM35 sensors, soil moisture sensors, and Firebase Cloud** for real-time environmental data collection across distributed sensor nodes. Built a **Python-based analytics and predictive automation pipeline** using Pandas and Scikit-learn to optimize irrigation and climate control workflows, improving crop yield by **20%**. Implemented cloud-based monitoring dashboards and automated data visualization pipelines using Matplotlib and Seaborn for operational analytics and remote monitoring.

Smart Parking Management SystemIoT | Flutter | Firebase | Cloud Computing

Developed a scalable smart parking management platform using **ESP32, Firebase Realtime Database, and Flutter**, enabling real-time parking slot tracking and mobile-based availability updates for 50+ concurrent users. Engineered a low-latency cloud synchronization pipeline with **<500ms response time** for real-time sensor-to-application communication, reducing average parking search time by **30%**. Implemented API-driven cloud communication and event-based data synchronization for reliable real-time updates.

Home Automation SystemIoT | Firebase | Flutter | AI Scheduling

Built a full-stack IoT-enabled home automation platform using **ESP32 microcontrollers, Firebase Cloud backend, and Flutter mobile frontend** to remotely manage and monitor 8+ connected appliances. Developed AI-based rule scheduling and automation workflows for smart energy optimization, irrigation management, and device control, reducing estimated energy consumption by **15%**. Designed secure cloud-connected communication workflows for real-time device synchronization and remote access.

Word/PDF Document ChatbotPython | NLP | Generative AI

Developed an **AI-powered document question-answering chatbot** using Python, NLTK, and vector-based semantic retrieval pipelines to support conversational querying across Word and PDF documents. Implemented semantic text chunking, NLP preprocessing, and intelligent retrieval workflows to process documents up to 100 pages while maintaining **sub-3 second response latency** and **85%+ answer accuracy**. Designed scalable document ingestion and contextual search pipelines for efficient natural language-based information retrieval.

EDUCATION

Bachelor of Technology in Computer Science with IoT, RGPV University (CGPA - 7.97)2021 - 2025

Senior Secondary Education (Class XII, 86.60%), State Board2020 - 2021

CERTIFICATIONS

Cloud Computing	(NPTEL)
Introduction to Cybersecurity	(Cisco)
CCNAv7: Introduction to Networks	(Cisco)
Communication Skills	(TCS iON)
Python Essentials	(Cisco)
Data Analytics	(Geeks of Gurukul)

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- **Competitive Problem Solving** – Solved **600+ Data Structures & Algorithms problems** on Geeks-forGeeks, covering Arrays, Linked List, Trees, Graphs, Dynamic Programming, and Recursion; demonstrating strong computational thinking and problem-solving skills.
- **Winner, SIStec Innovation Hackathon (SIH)** – Led a 5-member team in developing a cloud-integrated IoT solution for sustainable agricultural resource management, competing against 50+ teams.
- **Technical Speaker** – Delivered 3 presentations on Cloud Automation and IoT architecture to audiences of 50+ engineers and industry professionals.